

SAMUEL PATRONE

Theoretical Particle Physicist and Cosmologist

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📄 [iNSPIRE](#)

WHO I AM

I work at the crucial interface between abstract mathematical models and experimental observations in cosmology and particle physics. My research transforms theoretical predictions into testable signatures that can be probed in laboratories and telescopes. From strengthening axion-like-particle bounds to extracting fundamental physics from gravitational waves, I specialize in making the *invisible universe* experimentally accessible.

EDUCATION

Ph.D. in Physics

2020–2026

California Institute of Technology | GPA: 4.3/4.0

Koh Fellow (2020), Cullen Fellow (2021), Physics TA Fellow (2024-25)

Advisor: Prof. Mark B. Wise · Thesis: “Tracing New Physics: From Symmetry to Observation”

M.Sc. & B.Sc. in Physics

2015–2020

Sapienza University of Rome | 110/110 summa cum laude

Class Rank: #1 BSc (30.68/30), Top 5% MSc (30.32/30) · Sapienza School for Advanced Studies

European fellowships totaling €15,000+, including the Enrico Persico and INFN–NSF/LIGO fellowships

ACADEMIC APPOINTMENTS

Incoming Postdoctoral Scholar

Starting Sept. 2026

University of California, San Diego

Research in theoretical particle physics and cosmology with Prof. Aneesh V. Manohar.

Postdoctoral Scholar Research Associate

July–Aug. 2026

California Institute of Technology

RESEARCH IMPACT

Theoretical Particle Physics and Cosmology

2020–2026

Walter Burke Institute for Theoretical Physics (Caltech)

Studied warm inflation, leptogenesis in automatic Nelson–Barr models, galaxy-bias renormalization, axion-like particle production in beam-dump electromagnetic showers, and finite naturalness in quark–lepton unification. Combined analytical calculations and numerical simulations, yielding publications in PRD, JHEP, and JCAP.

Gravitational-Wave Phenomenology

2019–2020

LIGO Lab (Caltech) & Virgo (Sapienza Università di Roma)

Developed a Bayesian framework to extract gravitational-wave polarization content and test General Relativity. The resulting Master’s thesis received the 2020 Con.Sienze national prize.

INDUSTRY EXPERIENCE

Human Frontier Collective Specialist – Generative AI

May 2025–Present

Scale AI | part-time

Designed and validated complex physics problems – from graduate level to frontier – for training LLMs. Ensured scientific rigor of AI responses in physics, bridging my expertise with practical AI applications.

PUBLICATIONS & TALKS

Selected Publications:

- S. Patrone et al., “Long-lived axionlike particles from electromagnetic cascades,” *Phys. Rev. D* (2026)
- C. Murgui and S. Patrone, “Leptogenesis in automatic Nelson–Barr models,” *JHEP* (2025)
- P. Fileviez Pérez et al., “Finite naturalness and quark–lepton unification,” *Phys. Rev. D* (2024)
- S. Patrone et al., “Regularization-scheme dependence of the counterterms in the galaxy-bias expansion,” *JCAP* (2023)

As a member of the LIGO Scientific Collaboration and Virgo Collaboration from 2019 to 2021, I co-authored 10 additional peer-reviewed collaboration papers. A complete publication list is available on [iNSPIRE](#).

Invited Talks:

2026: UC Santa Cruz | 2025: Zurich, Texas A&M, Northwestern | 2024: CERN, Case Western

RECOGNITION

James A. Cullen Fellow

2021

Caltech academic excellence

Con.Sienze Prize

2020

National award for Master’s thesis research

Danny Koh Fellow

2020

\$18,000 graduate support

Enrico Persico Scholar

2020

€6,000 from Accademia dei Lincei

INFN/LIGO-NSF Fellow

2019

€5,000 for international research collaboration

TECHNICAL SKILLS

Theory: QFT, General Relativity, Cosmological Perturbations

Phenomenology: Cross Sections, Event Generation, Signal Analysis

Computing: Python, C, C++, Mathematics, ROOT

Methods: Adaptive Monte Carlo, Bayesian Inference, Boltzmann Equations, Loop Calculations

LANGUAGES

English (Fluent), Italian (Native), Spanish (Intermediate), Mathematics (Universal)

TEACHING & LEADERSHIP

Physics TA Fellow (2024-25)

Coordinated training and support for 100+ TAs; managed a 1.8K program budget and developed TA resources.

Teaching Excellence

100+ students, Certificate of Practice in University Teaching (Caltech)

Science Communicator (2018-20)

3 years bilingual museum tours (Sapienza)

Civic Leader (2015-18)

Youth Council President for 30K residents

BEYOND PHYSICS

Pianist · *Tiramisour*